ANALYTICAL CUMULATIVE EXAM

March 2, 1996

(60 points)

1. Molar absorptivities of absorbing molecules can be used to obtain compositional information for derivatives of the parent molecule (e.g. molecular weight) providing the added group does not affect the chromophore. For example, literature values for the molar absorptivities of nucleotides are often used to determine the concentration of nucleotide solutions, since even highly purified nucleotides may contain variable waters of hydration or a variable salt content. A sample of the disodium salt of cytidine 5'-monophosphate (5'-CMP, $C_9H_{12}N_3O_8PNa_2$) was weighed out (0.0814 g), dissolved and diluted to 25.00 mL in a volumetric flask with distilled water. ϵ for 5'-CMP is 1.30×10^4 at λ_{max} =280 nm in 0.01 M HCl 0.10 mL of the CMP stock solution was diluted to 10.00 mL with 0.01 M HCl. The absorbance of this solution was measured as 0.838 in a 1.0 cm cuvette.

a. Calculate the expected concentration of 5'-CMP assuming it to be a pure, anhydrous salt.

b. Calculate the concentration based on the spectrophotometric measurement.

c. Assuming the difference in concentration is due only to absorbed and bound water molecules, calculate the average number of water molecules per 5'-CMP molecule.

(40 points)

2. A new enzyme, named logicase has been discovered which has been isolated from the brain tissue of premed students. The reaction it catalyzes converts functioning cerebral tissue into repositories of useless and trivial facts. Those afflicted lose their ability to think logically but are able to memorize volumes of information. The reaction is given below:

Smartose	Logicase ---------->	Braindedic Acid
$[\alpha] = 45$		$[\alpha] = 10$
MW = 250		MW = 150

The units of $[\alpha]$ are deg dm^{-1} $(g/mL)^{-1}$. To study this reaction, a 0.15 g/mL solution of smartose was placed in the 1.0 cm cell of a polarimeter, the enzyme was added and the reaction allowed to proceed for 5 minutes. At the end of this time the solution had a rotation of 0.42 deg. Calculate the amount (g) of braindedic acid formed.